

Sagar Singh Bhola

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EDUCATION

University of Michigan—Ann Arbor

Bachelor of Science in Computer Science | **GPA: 3.75**

Ann Arbor, MI

Aug. 2021 – Dec. 2025

- **Coursework:** Data Structures and Algorithms, Computer Networks, Computer Security, Computer Architecture

TECHNICAL SKILLS

Languages/Technologies: Python, C/C++, JavaScript, TypeScript, SQL, Java, HTML/CSS

Frameworks/Libraries: React, Next.js, Vue, Node.js, Express.js, Flask, FastAPI, Pandas, PyTorch, OpenCV

Tools/Platforms: MySQL, Docker, AWS, Git, Postman, Playwright, Figma

EXPERIENCE

Data Engineering Co-op

Ciena Healthcare

May 2025 – Present

Southfield, MI

- Automated Michigan Medicaid cost-report preparation and validation across 51 nursing facilities, supporting an 84% first-pass acceptance rate and helping avoid approximately \$100K in external preparation costs.
- Built a Python ETL pipeline with Pandas, pdfplumber, regex, and OpenPyXL to process 2,000+ Medicaid reimbursement PDFs across four fiscal years, reducing a multi-day historical backfill to under two hours.
- Developed Python and SQL reconciliation pipelines comparing PointClickCare census and patient-in-bed data with general-ledger records, automatically flagging cross-system discrepancies before monthly financial close.

PROJECTS

Vision-Based Autonomous Driving Agent | *Python, PyTorch, C#, ONNX, DirectInput*

Aug. 2026 – Present

- Built a vision-based driving model in PyTorch that predicts steering, throttle, and braking from camera images and vehicle speed using behavior cloning.
- Collected and synchronized gameplay images with vehicle telemetry, split recordings by session to avoid data leakage, and achieved a steering MAE of 0.046 on a separate test session.
- Connected the trained model to a Thrustmaster T300RS through C#, ONNX Runtime, and DirectInput, applying live steering commands with approximately 25 ms processing latency.

AI Invoice Processor | *Python, Next.js, FastAPI, Ollama, Qwen, Docling*

Apr. 2026 – May 2026

- Built a privacy-focused invoice processor that converts uploaded documents into structured accounting data using a locally hosted language model and schema-constrained extraction.
- Developed an end-to-end workflow for invoice ingestion, extraction, validation, human review, and persistence, reducing average entry time from 4 minutes to 30 seconds.
- Implemented stateful accounts-payable review with embedded PDF previews, editable invoice fields and line items, and persistent drafts for human-in-the-loop correction.

YesNoFingers | *Python, OpenAI Whisper, AWS Translate, AWS EC2*

May 2025 – Present

- Built a cloud-based medical speech-translation system with the University of Michigan Language Resource Center, supporting real-time translation across 40+ languages using Whisper and AWS Translate.
- Engineered a Python pipeline on AWS EC2 for audio validation, transcription, terminology-aware preprocessing, translation, and API responses, achieving approximately three-second end-to-end processing per recording.
- Piloted the system with five clinicians at a Detroit-area free clinic, using their feedback to improve API reliability, input validation, and failure handling.

ACHIEVEMENTS

Cyber Threat Competition - Finalist, Technology Lead

Deloitte Risk & Financial Advisory

Feb. 2023

Westlake, TX

- Selected as 1 of 60 finalists from 30 universities after completing cybersecurity questionnaires and CTF challenges; advanced to the live finals at Deloitte University.
- Led technical analysis during a two-day cyber incident response simulation and presented response recommendations to a panel of Deloitte Risk & Financial Advisory leaders.